

Enhancing user experience in Virtual Reality Through Optical Flow Simplification with the help of Physiological Measurements: Pilot Study

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Abstract

The use of virtual reality (VR) has made significant advancements, and now it's widely used across a range of applications. However, consumers' capacity to fully enjoy VR experiences continues to be limited by a chronic problem known as cybersickness (CS). This pilot study explores the feasibility of mitigating cybersickness through geometric scene simplification combined with EEG-based monitoring. According to the sensory conflict theory, this issue is caused by the discrepancy between the visually induced self-motion (VIMS) through immersive displays and the real motion the vestibular system detects. While prior mitigation strategies have largely relied on hardware modifications or visual field restrictions, this paper introduces a novel framework that integrates geometric scene simplification with electroencephalography (EEG) based neurofeedback to actively reduce VIMS during VR immersion. The proposed framework combines EEG neurofeedback, allows monitoring users' brain wave activity and cognitive states during virtual immersion experience. The empirical evidence from our investigation shows a correlation between CS manifestation and neural activation in the parietal and temporal lobes. As a pilot experiment with a limited sample, statistical differences were not significant; however, preliminary trends suggest lower neural activation during simplified scenes. Notably, a decrease in neural activation corresponding to reduced optic flow (OF) suggests that VR environment simplification may help attenuate CS symptoms, providing preliminary support for the proposed strategy.

Keywords: Optical flow, cybersickness, virtual reality, electroencephalography

1. Introduction

VR technology has witnessed rapid advancements, leading to its widespread adoption in various domains such as gaming, training, and medical applications [1]. Despite its potential, one of the significant challenges holding back the full utilization of VR is CS [2]. CS is a phenomenon that shares similarities with motion sickness [3]. Interestingly, CS can be triggered solely by visual stimuli, in the absence of corresponding physical movement. The heightened visual stimulus in such environments provides users with a greater wealth of environmental information, making it more challenging to ignore the conflicting sensory cues. Previous study have explored various factors contributing to CS, including visual vestibular conflicts, display latency, and user susceptibility [4].

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In contrast to prior work that has mainly relied on hardware adjustments [5], visual field restrictions [6,7], or post-processing [8], the novelty of our study lies in combining geometric simplification of virtual scenes with EEG-driven analysis of optic flow. This integration enables both objective detection of cybersickness markers and dynamic adjustment of visual complexity to mitigate symptoms. To contextualize this contribution, it is important to note that CS is commonly studied through user evaluations, since it is inherently a subjective experience. Some methods are intrusive, like using sensors to track body signals [9], while others rely on what participants say during or after the session, which may not always be accurate [10]. Our study suggested an alternative way to distinguish CS is by analyzing visual features like optical flow (OF), which tracks motion between video frames to understand visual movement. A study by Smith [11], introduced a metric combining OF and entropy to assess the potential of VR environments to induce CS. The findings indicated that environments with higher OF entropy were more likely to cause CS symptoms. This suggests that increased complexity and intensity of visual motion contribute to discomfort during VR interaction. The limitation of his work is that OF data were extracted from a stationary viewpoint, thus capturing only the environmental motion while excluding user-induced head movements that typically contribute to CS. Although widely researched in VR, OF has some limitations. It includes horizontal, vertical, and magnitude components, which all affect the viewer differently but happen at the same time, making it hard to study separately [12,13]. Also, the amount of optical flow can vary depending on how fast the user moves or how the environment is designed, and a high number of optical flow events don't always mean the experience will cause CS [11]. To address the limitations of relying solely on OF for predicting CS, pilot study proposes a new approach to objectively characterize self-motion experiences in head-mounted display (HMD) virtual environments (VE). The method aims to identify CS inducing features as part of an iterative design process, allowing for the evaluation and refinement of virtual scenes before user testing. It combines OF metrics with an analysis of visual motion in both central and peripheral fields of view (FOV). This approach also incorporates scene simplification in the peripheral view and compares simple versus complex environments, alongside the use of physiological signals to enhance prediction accuracy. One of the well-known neurophysiological measurements is EEG, which can be used to investigate the neural correlations of CS, offering insights into the brain's response during VR exposure [14,15]. For instance, changes in EEG power spectral densities in specific frequency bands have been associated with the onset of CS symptoms [16,17]. Several approaches have been proposed to mitigate CS, such as adjusting FOV restrictions, incorporating rest frames, and employing motion prediction algorithms [18,19]. While there are practices that can alleviate CS within VE, such as narrowing the FOV [7] or incorporating background images [20], these methods tend to reduce the sense of presence and immersion, thereby continuing to hinder the widespread adoption of VR. To address these challenges, it becomes imperative to employ effective strategies for the enhancement of visual perception in VR to mitigate the CS.

CS detection can be achieved by analyzing physiological signals collected via sensors such as Electrocardiogram (ECG) and Galvanic Skin Response (GSR) [21]. Neurophysiological devices employing GSR, EEG, and Photoplethysmography (PPG) techniques are used to obtain accurate, noninvasive insights within psychiatry. Real-time measurements and reduction of CS with the help of EEG measurement have also shown promising outcomes [22]. However, there remains a need for more effective and objective methods to detect and alleviate CS [23,24].

Subjective self-reports can be used to assess the presence of VR induced symptoms and effects (VRISE). The Simulator Sickness Questionnaire (SSQ) [25], which has become the standard tool to quantify VRISE in VR, is the most often used subjective questionnaire

for assessing VR-induced symptoms and effects. SSQ has been utilized in various works [26,27]. However, not many studies have introduced physiological measurement methods, for instance fNIRS or EEG.

EEG is widely used in brain-machine interface (BMI) applications due to its affordability, high temporal resolution, and compatibility with diverse devices. EEG provides significant advantages over other brainwave measurement techniques, such as electrocorticography (ECoG) and magnetoencephalography (MEG). Primarily, EEG is non-invasive, eliminating the need for electrode insertion into the brain [28]. A study by Krokos [15] recorded EEG during a Vection inducing VR fly-through and compared CS against baseline. They reported significant increases in delta (1–4 Hz), theta (4–7 Hz), and alpha (7–13 Hz) power correlating with joystick reported sickness. This supports EEG as a continuous, real-time marker, in contrast to self-reported questionnaire data. A study by Hasan [29] emphasizes the critical importance of assessing cybersickness through tools like the Simulator Sickness Questionnaire (SSQ) and evaluating presence metrics to gauge user discomfort and engagement levels in VR environments. Recent work by Demirel [30] proposed a real time continuous estimation method combining dry electrode EEG + head motion, using multitaper spectral analysis and Long Short-Term Memory (LSTM) algorithm, which tracks dynamic CS levels without calibration, overcoming limitations of post VR questionnaires like SSQ and enabling more granular tracking. These works demonstrate the potential of EEG for dynamic monitoring, but they stop short of coupling detection with adaptive VR design. By combining EEG analysis with geometric simplification and optic flow adjustments, the present study addresses this gap, offering both objective detection and proactive mitigation within immersive VR environments.

In VR discrepancies between visual motion cues and physical movement can disrupt perception, leading to discomfort. FOV plays a significant role in this context; These OF patterns have the potential to induce pronounced Vection [31] which contradicts the motion cues perceived by the vestibular system. Reducing FOV can mitigate motion sickness by limiting peripheral motion cues, yet it may also diminish immersion [6]. To address this, techniques like dynamic FOV adjustment have been proposed, which adaptively narrow the FOV during rapid movements to balance comfort and immersion [32]. Another strategy involves implementing rest frames stable visual references within the VR environment, such as a virtual nose or cockpit which help anchor the user's perception and reduce sensory conflict [33], while this strategy can alleviate CS, this often does so at the expense of immersion by restricting the user's view or adding artificial anchors. The proposed approach differs by aiming to reduce cybersickness without compromising immersion, through geometric simplification of scene complexity guided by EEG based monitoring. Innovative approaches like peripheral teleportation create a stable peripheral view aligned with the user's physical motion, further mitigating CS while preserving immersion [34]. These findings underscore the importance of carefully designing visual cues and reference frames in VR to enhance user comfort and experience. A study conducted by [35] employed geometric simplification to diminish optic flow, aiming to mitigate CS in the context of VR, also the same author proposed the use of Geometric deformation would reduce the optics flow causing less CS in VR environment [36], The study did not incorporate EEG, eye-tracking, or physiological indicators e.g., heart rate, skin conductance, which could have provided deeper insight into the sensory conflict and its neural processing. In contrast, our work advances this line of research by combining scene simplification with EEG-driven analysis, enabling both detection and mitigation of cybersickness. Another study by [37] suggested that mitigating CS in VR could be achieved through the implementation of Adaptive FOV Limitations, which restrict the OF experienced by users. Consequently, the phenomenon of CS in VR can be attributed to the OF resulting from locomotion, particularly when users engage in virtual movement without corresponding

physical motion. In the context of wide FOV displays. Geometric mismatches in FOV restriction may lead to suboptimal occlusion or undesirable visual artifacts. On the other hand, proposed method avoids such artifacts by directly simplifying the geometry of the virtual environment, allowing a more natural reduction of optic flow that aligns with user perception. Given the exploratory nature of this work and the limited number of participants, the present study should be regarded as a pilot investigation aimed at validating the feasibility of integrating EEG feedback with geometric simplification to mitigate cybersickness.

2. Materials and Methods

To investigate the neurophysiological correlations of CS induced by varying levels of OF, three geometrically distinct VE were developed using Unity 3D. These environments included: (1) a high OF scenario featuring complex visual details, (2) a moderately simplified scene with a 50% reduction in geometric complexity, and (3) a fully simplified scene with a 100% geometric reduction (Figure 2, first row).

Upon wearing HMD, participants were immersed in a street-like VE populated by animated zombies and surrounding buildings. Their task was to shoot at the zombies while navigating the VE. This interactive task was consistent across all three visual complexity conditions. EEG data were recorded throughout the exposure to analyze the relationship between OF intensity and neural responses. The experimental design enabled the evaluation of how increasing visual complexity and OF levels influenced CS symptoms and corresponding EEG patterns.

The interactive zombie-shooting task was intentionally included to maintain a consistent level of attention and engagement across all conditions. Importantly, the gameplay itself did not vary between scenes; only the geometric complexity and corresponding optical flow were manipulated. This controlled design ensured that any observed differences in EEG activation or cybersickness trends could be attributed primarily to variations in visual motion complexity rather than to changes in task difficulty, workload, or emotional arousal.

PARTICIPANTS

This study engaged four healthy male participants, aged from 25 to 40 years old (32.75 ± 6.34). While the sample size is limited, this pilot study aimed at exploring conceptual feasibility and analyzing initial EEG and CS-related data in VR [38–40]. All the participants are free from ophthalmologic conditions, following the acquisition of their written informed consent. Prior to the experiment, participants abstained from consuming substances such as alcohol, drugs known to induce nausea or headaches, and caffeine. To ensure the integrity of EEG data, participants were instructed to minimize head movements and remain silent during the EEG recording sessions, which were challenged by some technical difficulties associated with EEG data acquisition. This study was approved by the Ethics Committee of the University Pendidikan Sultan Idris (No. UPSI/PPPI/PYK/ETIKA(M)/Jld. 19(89)).

APPARATUS

EEG signals were recorded using the eego™ sports EE-201 amplifier (AntNeuro, eemagine Medical Imaging Solutions GmbH, Gubener Str., Germany). Electrode placement followed the international 10–20 system [41], with electrodes positioned as shown in (Figure 1). A forehead placed at the Ground (GND) electrode, while the reference electrode was located at CPz. All electrode impedances were carefully maintained below 10 kΩ to ensure signal quality. During the experiment, participants wore an Oculus Quest 2 VR headset.

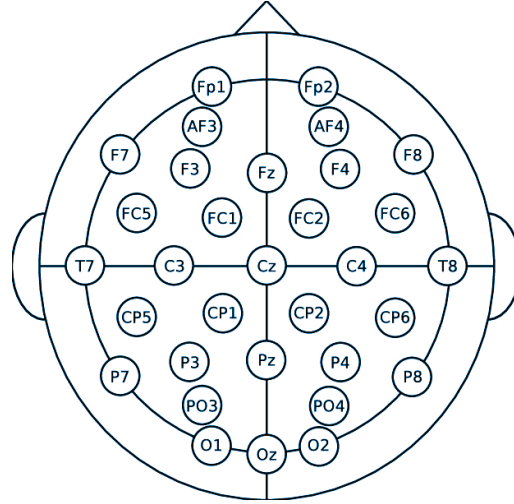


Figure 1. International 10-20 system for 32 channels connection [42].

Using EEG, brain activity will be recorded during each scenario. Preprocessing steps will remove artifacts, followed by feature extraction using methods such as Independent Component Analysis (ICA), complex Morlet wavelet transforms, and power spectral density (PSD) analysis to identify time-frequency patterns associated with CS.

PRE-PROCESSING

EEG data were pre-processed using the open-source EEGLAB toolbox in MATLAB, with a sampling rate of 512 Hz. To attenuate slow drifts and high-frequency noise, a band-pass finite impulse response (FIR) filter was applied with cutoff frequencies of 1 - 40 Hz, using a Hamming windowed sinc filter. Power line noise at 50 Hz was removed using the CleanLine plugin, applied to ICA components using a sliding window (4 s length, 1 s step) and a spectral smoothing factor of 100. Noisy or corrupted EEG channels were visually identified and interpolated using spherical spline interpolation, only when clearly necessary. The data were then re-referenced to the average of all channels to minimize spatial noise. Finally, Independent Component Analysis (ICA) was performed to identify and remove artefactual components related to eye movements and muscle activity. All pre-processing procedures were consistently applied across all participants to ensure data quality and comparability.

GEOMETRICAL SIMPLIFICATION AND OPTICAL FLOW ANALYSIS

To investigate the impact of geometrical complexity on user discomfort in VR, three distinct scene conditions were developed as shown in Figure 2. Each scene varied in terms of geometry complexity, texture detail, thereby influencing the amount of OF perceived during navigation Figure 2 (second row).

The OF measured using Gunnar Farneback algorithm [43]. Farneback approximates the neighbourhood of each pixel in an image as a quadratic polynomial:

$$f(x) \approx x^T A x + b^T x + c \quad (1)$$

Where $x=(x,y)$ pixel position, A is a symmetric matrix second order derivative, and b is a vector first order derivative and c is a scalar offset. The coefficients are estimated from a weighted least squares fit to the signal values in the neighborhood.

Suppose between two frames I_1 and I_2 , a small neighborhood is translated by a flow vectord. Then the polynomial coefficients change accordingly:

$$f_2(x) = f_1(x - d) \quad (2)$$

So, we compare polynomial expansions of f_1 and f_2 and the goal is to find the displacement d that best aligns the neighborhoods (minimizing error between polynomials). The optimal displacement vector d is obtained by minimizing the difference in local polynomial representations between frames:

- A_1, b_1, c_1 from frame 1,
- A_2, b_2, c_2 from frame 2.

We then solve:

$$d = \arg \min \int_{x \in \Omega} [f_1(x) - f_2(x + d)]^2 dx \quad (3)$$

Farneback's algorithm applies this displacement estimation densely across the image using: Image pyramids (multi-scale), Smoothing and filtering (Gaussian), Iterative refinement. This gives a dense flow field:

$$flow(x, y) = (u(x, y), v(x, y)) \quad (4)$$

Where u and v are the horizontal and vertical displacements respectively for each pixel.

The direct application of the Euclidean norm (L2 norm) is used to measure the magnitude of a vector as follows:

$$Magnitude = \sqrt{(dx^2 + dy^2)} \quad (5)$$

The weighted arithmetic means a standard statistical measure:

$$Weighted\ Avg = \frac{\sum(mag \cdot weight)}{\sum(weight)} \quad (6)$$

Normalization and scale value to a relative percentage:

$$Percent = \frac{avg_{mag}}{max_{flow}} \times 100 \quad (7)$$

The OF is computed using Farneback's dense, OF algorithm from OpenCV using the following python library called `cv2.calcOpticalFlowFarneback`. The Complex condition, representing a fully detailed scene without any geometric simplification Figure 2 (a), exhibited the highest OF at 70.1%. This scenario simulates a visually rich and dynamic environment, which is commonly linked to elevated levels of motion sickness due to increased visual complexity and motion cues. In contrast, the repetitive wall designs, significantly reducing OF to 21.7% and 16.4%, respectively. This simplification aims to minimize the perceptual load and reduce sensory conflict. 50% and 100% Simplified conditions Figure. 2 (b) and 3(c) eliminate architectural textures and introduce uniform, these variations in OF directly correlate with user discomfort, as summarized in Table 1. The Complex scene recorded the highest OF scores (mean: 6.23, median: 6.55). The simplified conditions show a substantial decline in OF. For instance, the Simplified 50% scene report average OF scores of 2.02. The Simplified 100% condition further decreases OF (mean: 1.95) along with flow percentage (16.4%). Reducing visual motion through geometric sim-

Condition	Mean	Median
complex	6.23	6.55
simplified 50%	2.02	1.85
simplified 100%	1.95	1.74

Table 1. Optical Flow Means and Median Average

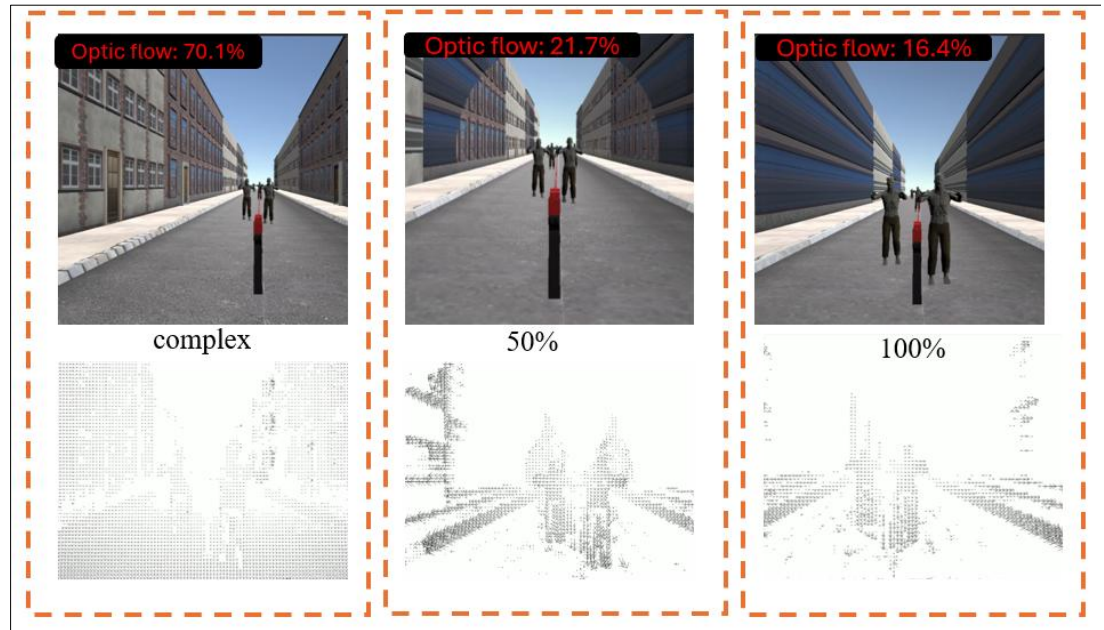


Figure 2. VR scene showing optical flow simulation: Left column: complex scene with the highest optical flow, Middle column: simplified scene with 50% reduction, and Right column: fully simplified scene with the least optical flow (100% simplification).

plification and the removal of dynamic stimuli is shown to significantly alleviate symptoms of CS, making it a valuable design consideration for enhancing immersive VR experiences [44,45].

DATA COLLECTION PROTOCOLS

After confirming the validity and reliability of the experimental protocol and obtaining ethical consent, EEG was integrated with VR technology. The experimental procedure commenced with a baseline recording of 120 seconds, followed by the first task (complex scene) of 25 seconds duration, a subsequent rest period of 10 seconds, a second task (50% simplified scene) lasting 25 seconds and same for last task (100% simplified scene), and a final rest interval of 10 seconds. This sequence was repeated across 5 trials, culminating in total duration of approximately 17.5 minutes, also post baseline record also taken to further study the different brain activity before and after the interaction with VR environment. The relatively short exposure duration (25 seconds per scene) was deliberately chosen to minimize participant discomfort and prevent severe cybersickness during this pilot study. Longer exposure times will be implemented in future experiments to better capture the gradual onset of cybersickness symptoms. Figure 3 illustrates the timeline for all the scenes. Prior to the commencement of EEG recording, participants underwent training to familiarize themselves with the VR environment, including trial runs. Participants were instructed to sit comfortably and minimize blinking and movement as much as feasible during the EEG recording sessions to ensure the accuracy and reliability of the data collected. All procedural conditions, alongside potential effects, and the participants' autonomy to withdraw at any moment, were thoroughly communicated. The entire experimental session, inclusive of EEG electrode placement, was conducted within a time frame of less than one hour.

3. Results and Discussion

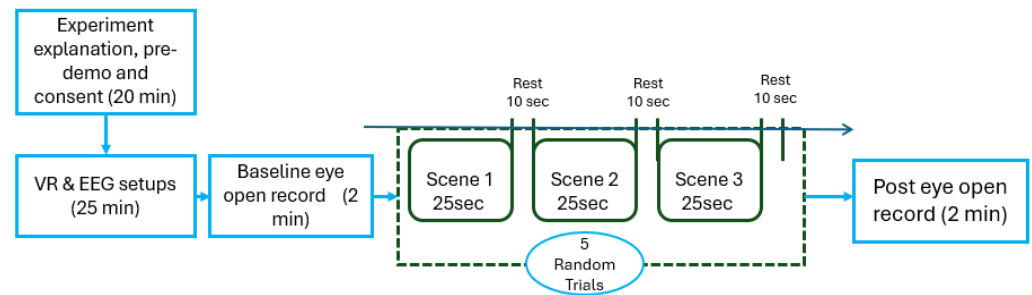


Figure 4. Data Collection Timeline and The Number of Trials.

The following findings represent preliminary observations based on a small participant cohort. They should be interpreted as feasibility indicators rather than statistically generalizable outcomes.

3.1 Brain Topography

Figure 4. presents the EEG spectral power topography across four frequency bands (delta, theta, alpha, and beta) for three experimental conditions: condition 1 (100% simplification), condition 2 (50% simplification), and condition 3 (complex scene). When analysing these EEG bands, a notable observation is that lower amplitude values, which are closer to baseline levels, are typically associated with fewer symptoms of CS. This trend appears most pronounced in condition 1, where reduced power density in the parietal and occipital areas for the Delta, Theta, and Beta waves indicate a reduction in the participant's susceptibility to CS. This reduction in neural activity suggests that less visually stimulating conditions, such as those in condition 1, evoke fewer symptoms related to motion sickness.

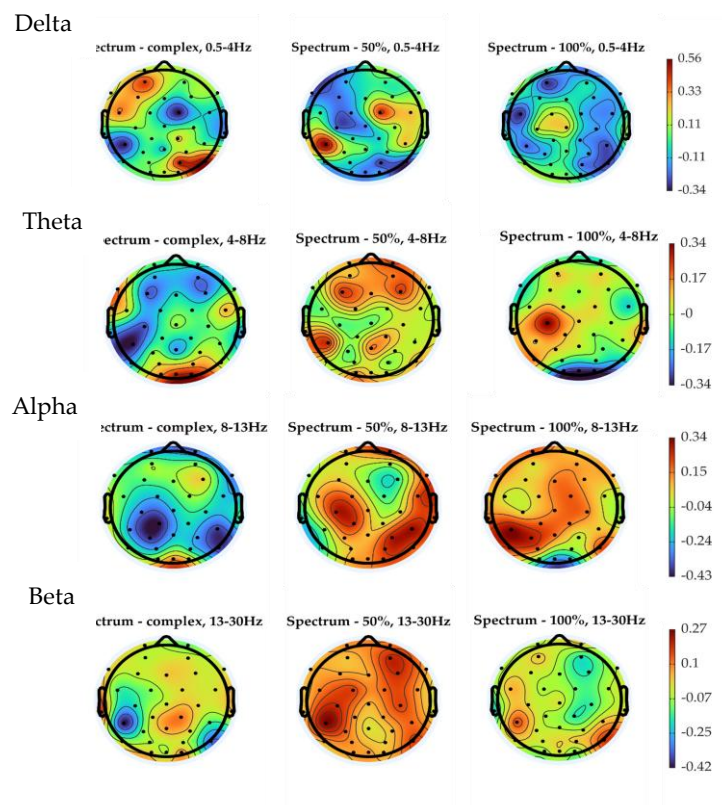


Figure 3. Brain Topography results in the complex, 50% and 100% simplification.

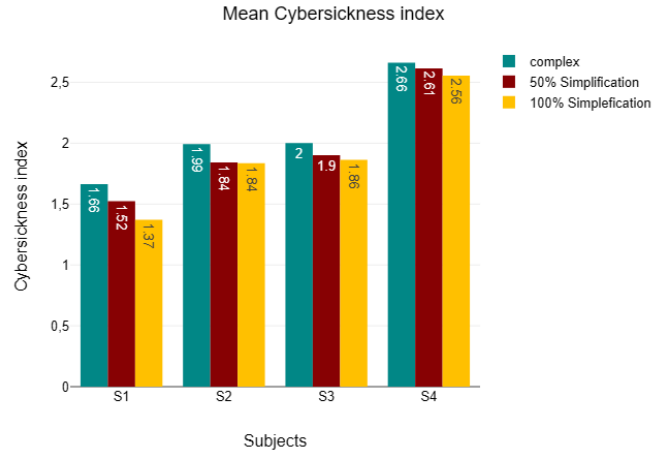


Figure 5. EEG Waves-Based Condition and Averaged PSD for Four Subjects.

Conversely, condition 3, which represents a more complex scene, seems to be associated with greater amplitude fluctuations across multiple regions, particularly in the Delta, Theta, and Beta bands. These fluctuations suggest a stronger neural response to more complex visual stimuli, which correlate with increased symptoms of CS. The greater variability in power density reflects an overstimulation of the sensory system, which is often linked to discomfort and disorientation in VE. Regarding the alpha wave band, higher levels of alpha power have been widely associated with relaxation and reduced sensory input processing. The elevated alpha power in certain conditions, such as those seen in condition 1, indicates a more relaxed or comfortable state for participants. This observation aligns with established research showing that increases in alpha power are linked to states of relaxation and comfort [46–49]

3.2 Statistical Analysis and Results

Figure 5 visualizes the CS ratings across four subjects (S1 to S4) under various conditions, including different levels of scene simplification and environmental factors. Preliminary observations indicate that Subject 4 (S4) consistently exhibited the highest activation levels across all conditions, suggesting heightened susceptibility to CS. Across subjects, the complex scene appeared to evoke stronger CS symptoms than the simplified conditions, with the 100% simplification condition consistently showing the lowest CS ratings. While these patterns align with prior EEG studies [15,50], they should be interpreted cautiously given the limited sample size, and considered as initial evidence supporting the potential of geometric simplification to alleviate CS. The table 2 and Figure 6 shows the power values across different EEG frequency bands (delta, theta, alpha, beta, and gamma) for 3 scene conditions.

The key focus of this analysis lies in the role of alpha, delta, theta, and beta waves in relation to CS, as the Gamma band remains unchanged across all scenes and has thus been

Scenes	Delta	Theta	Alpha	Beta	Gamma	Total
Complex	0.63	0.64	0.67	0.45	0.06	0.49
50% Simplification	0.62	0.66	0.7	0.46	0.06	0.51
100% Simplification	0.61	0.64	0.74	0.46	0.06	0.5
Resting before task	0.7	0.71	3.74	0.83	0.07	1.21
Resting after task	0.72	0.74	5.34	0.9	0.09	1.56

Table 2. EEG Waves-Based Condition and Averaged *PSD* for All Subjects

excluded from our CS index. This omission is consistent with prior research, where the gamma band has similarly been found to exhibit minimal variation during conditions associated with CS [51]. A significant observation is the behavior of the alpha band, particularly in the more complex scenes. Alpha power shows marked suppression during the "complex" condition (0.67). This pattern of alpha suppression during vection or high sensory conflict is consistent with prior findings [49]. In this context, the suppression of alpha power indicative of heightened neural engagement, possibly explaining the increased CS reported in more complex VE. Alpha power exhibits a significant increase during rest states, particularly "Rest After" (5.34), where it peaks. This elevation in alpha power during rest is consistent with the association of alpha waves with relaxation and reduced sensory input processing, further supporting the idea that higher alpha power corresponds to a more comfortable and relaxed state for participants.

Four tests (Kolmogorov-Smirnov, Smirnov with Lilliefors correction [52], Shapiro-Wilk [53], and Anderson-Darling [54]) produced p-values greater than 0.05 as shown in table 4, indicating that there is no significant deviation from normality. Therefore, it is appropriate to proceed with statistical methods that assume the normality of the data based on these test results.

A mixed-model ANOVA was conducted to examine the effects of EEG frequency band (delta, theta, alpha, beta, gamma) and scene condition (Complex, 50% Simplification, 100% Simplification, Resting Before Task, Resting After Task) on EEG power. The analysis revealed a significant main effect of frequency bands, $F(4, 60) = 11.57$, $p < .001$, indicating that EEG power significantly differed across bands. There was also a significant main effect of condition, $F(4, 24) = 2.63$, $p = .036$, suggesting that the visual scene conditions had a notable impact on EEG responses. The effect size for this treatment effect, as measured by eta squared (η^2), was 0.02, indicating that approximately 2.23% of the variance in EEG power can be explained by differences between the conditions. This represents a small to medium effect size. Moreover, a significant interaction between band and condition was observed, $F(16, 60) = 2.5$, $p = .001$, implying that the influence of scene condition on EEG power varied depending on the frequency band. Notably, beta power was highest during the "Resting before task" condition. In contrast, a two-way ANOVA on CS scores across the same conditions and five trials showed no significant main effects of condition ($p = .302$) or trial ($p = .991$), and no significant interaction ($p = .324$). The post-hoc comparisons (Bonferroni, Tukey, and Scheffé) show that none of the differences between the three conditions (100%, 50%, Complex) were statistically significant ($p > 0.05$).

This lack of significant differences suggests that neither the tasks nor the number of trials had a substantial impact on CS, at least within the observed conditions. However, it is essential to note that averaging across multiple trials is a more robust method for reducing variability and increasing the accuracy of the data. Single-trial measurements can

Statistical Tests	Complex		50% Simplification		100% Simplification	
	Statistics	p	Statistics	p	Statistics	p
Kolmogorov-Smirnov	0.31	.747	0.26	.875	0.33	.658
Kolmogorov-Smirnov (Lilliefors Corr.)	0.31	.268	0.26	.637	0.33	.146
Shapiro-Wilk	0.92	.526	0.95	.743	0.88	.351
Anderson-Darling	0.32	.311	0.26	.502	0.38	.21

Table 3. Tests of Normality Across Scene Conditions: Kolmogorov-Smirnov, Shapiro-Wilk, and Anderson-Darling Results for Each Level of Geometric Simplification.

Conditions	MWI	CS	Total
100% Simplification	0.91	1.92	1.42
complex	0.98	2.06	1.52
50% Simplification	0.92	1.96	1.44
Resting Before VR	0.27	1.25	0.76
Resting after VR	0.13	1.23	0.68

Table 4. CS, mental workload index (MWI), and tress score for six conditions

be prone to outliers or random fluctuations in participant responses, especially in subjective experiences such as CS. By conducting multiple trials and taking the average, researchers can obtain a more reliable estimate of participants' typical experience. Hence, even though no significant differences were found in this analysis, using multiple trials still offers a more valid assessment of the phenomena being studied.

Table 4 and Figure 7 present average data across all trials, providing insight into the levels of CS, mental workload, across different experimental conditions. The results show that the CS levels are highest in the "complex" scene (mean = 2.06), while the lowest levels were observed during rest states, particularly "Rest After" (mean = 1.23). This reinforces the understanding that more visually demanding scenes, like the complex environment, tend to induce greater symptoms of SC compared to simpler scenes or resting states.

When analysing the mental workload and CS, it becomes evident that participants experienced the highest levels of mental workload in the "complex" condition (mean = 0.98) and "50% Simplification" (mean = 0.92) as shown in Figure.7. b, which corresponds with the elevated CS scores.

When considering the three VR conditions along with resting-state measures before and after the game, the mixed model ANOVA confirmed that there was a significant difference between the different groups for Mental Workload, and CS, with $F(1, 15) = 222.54$, $p < .001$, indicating that these variables significantly influence the dependent variable. Additionally, there was a significant difference between the resting state and the three conditions, $F(4, 15) = 5.81$, $p = .005$, suggesting that the different experimental conditions affected participants' CS, and workload levels.

Importantly, while there was a significant difference between the resting states and the other conditions, there was no significant difference between the experimental conditions themselves, such as between the "complex" and "100% simplification" scenes ($p = 0.91$). This indicates that while rest states stand out for reducing symptoms of CS, the dif-

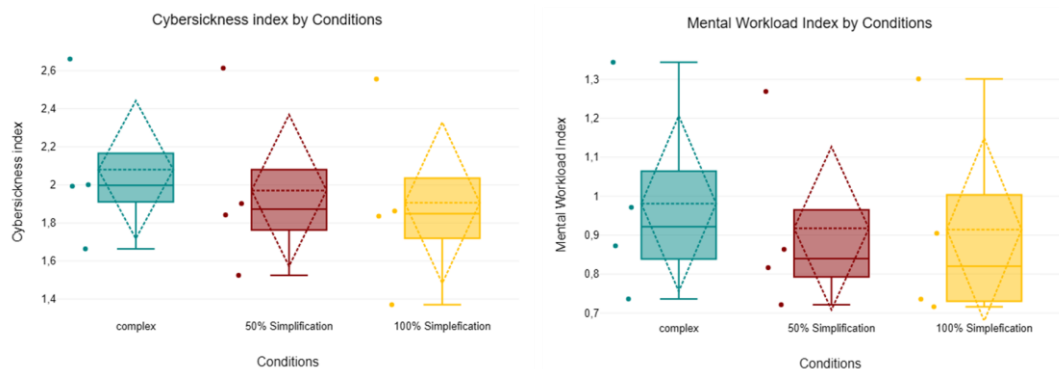


Figure 6. Averaged Power Density for The CS and Mental Workload Index.

ference between varying levels of scene complexity does not appear to significantly impact these metrics.

The absence of statistically significant differences between the experimental conditions, such as the “complex” and “100% simplification” scenes ($p = 0.91$), can largely be attributed to the limited sample size of this study ($n = 4$), which limits statistical power and makes it difficult to detect condition specific effects. Nevertheless, despite the lack of statistical significance, consistent patterns were observed across subjects, with simplified scenes generally associated with lower CS ratings. These trends, while preliminary, offer valuable insights into the potential benefits of scene simplification and underscore the importance of further investigation with a larger participant pool.

3.3 Limitations

While this study presents a promising approach to mitigating CS through geometric simplification and EEG-based neurofeedback, several limitations must be acknowledged. The most significant constraint is the limited sample size, with only four participants included in the pilot study. This small cohort reduces the statistical power of the findings and limits the generalizability of the results. Consequently, some observed differences between conditions such as between the complex and simplified scenes did not reach statistical significance, despite apparent trends.

The experimental conditions themselves were also relatively short (each scene lasting only 25 seconds), which may not fully reflect the effects of prolonged VR exposure that often intensifies CS. Furthermore, the study focused mainly on geometric simplification, but other contributing factors such as texture, color dynamics, and interactive elements like zombies were not fully isolated or controlled for. Additionally, the zombie-shooting task may have introduced stress, attention, and excitement effects that could influence EEG signals independently of CS.

3.4 Future Works

Increasing the sample size to enhance statistical robustness and enable more definitive conclusions about condition specific effects. Extending the exposure duration to better simulate realistic VR usage and observe long term impacts on CS and brain activity. Also incorporating multimodal physiological sensors e.g., GSR, PPG alongside EEG to provide a more comprehensive picture of user discomfort and cognitive load. Exploring machine learning models to predict CS in real time based on EEG features, allowing for adaptive VR content delivery.

4. Conclusions

In conclusion, this pilot feasibility study explored discomfort mitigation in VR environments through a framework that integrates EEG neurofeedback with geometric simplification. This approach allowed monitoring and analysis of user’s cognitive states during different VR conditions. Preliminary observations indicate a relationship between CS and increased neural activity, particularly in the parietal and temporal lobes, and suggest that geometric simplification may help reduce neural activation associated with CS. Trends also indicate a correlation between higher optic flow in the scene and elevated CS levels. While these findings are not statistically conclusive due to the small sample size, they provide exploratory evidence supporting the potential of combining EEG-based monitoring with optic flow simplification as a novel approach to CS mitigation. This study contributes new insights into the design of VR experiences that aim to balance immersion and user comfort. For future work, expanding the sample size and diversity will be essential to validate the preliminary findings and strengthen statistical significance. Moreover, investigating adaptive real-time geometric simplification strategies based on individual user responses could lead to more personalized and effective VR discomfort reduction.

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Abbreviations

The following abbreviations are used in this manuscript:

VR	Virtual Reality
CS	Cybersickness
FOV	Field Of View
OF	Optical Flow
VRISE	Virtual Reality induced symptoms and effects
SSQ	Simulator Sickness Questionnaire
HMD	head-mounted display

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